

Lecture 25

$A = A^T$

- $\lambda \rightarrow$ all real
- eigenvectors $\rightarrow$ all orthogonal
can be chosen

$A = S \Lambda S^{-1}$

and for symmetric $\Rightarrow$

$$A^T = S \Lambda S^{-1}$$

$\dots \rightarrow$ can be chosen $\perp$ and unit.

$A = Q \Lambda Q^{-1}$

$$A = Q \Lambda Q^T$$

$$\text{as } \underline{Q^{-1} = Q^T}$$



Spectral theorem

Why real eigenvalues

$$A x = \lambda x \Rightarrow \bar{A} \bar{x} = \bar{\lambda} \bar{x}$$



complex conjugate

For real A ,

$$\Rightarrow A x = \lambda x \Rightarrow \bar{x}^T A x = \bar{\lambda} \bar{x}^T x$$

$$\bar{x}^T A^T = \bar{\lambda} \bar{x} \Rightarrow x^T A = \bar{\lambda} \bar{x}$$

Now we can see
Assuming $\bar{x}^T x \neq 0$

$$\boxed{\lambda = \bar{\lambda}}$$

only for real.

$$\Rightarrow \bar{x}^T x = [\bar{x}_1 \quad \bar{x}_2 \quad \dots \quad \bar{x}_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$\Rightarrow \bar{x}_1 x_1 + \bar{x}_2 x_2 + \dots + \bar{x}_n x_n$$

$$\boxed{(a+ib)(a-ib) = a^2 + b^2}$$

So, x would have to be a zero vector for $\bar{x}^T x$ to be zero

For our proof, $\bar{A}^T = A$ is a requirement.

$$A = Q \Lambda Q^T$$

$$\begin{bmatrix} | & | & & \\ q_1 & q_2 & \dots & \\ | & | & & \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} -q_1^T \\ -q_2^T \\ \vdots \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} | & | & & \\ \lambda_1 q_1 & \lambda_2 q_2 & \dots & \\ | & | & & \end{bmatrix} \begin{bmatrix} -q_1^T \\ -q_2^T \\ \vdots \end{bmatrix}$$

$$\Rightarrow A = \lambda_1 \underbrace{q_1 q_1^T}_{\text{projections}} + \lambda_2 q_2 q_2^T \dots$$

→ Every symmetric matrix is a combination of perpendicular $\perp$ projection matrices

Neat fact !!

The # signs of the pivots for symmetric matrices are same as # signs of eigenvalues.

Positive Definite Matrix

- Symmetric
- All +ve eigenvalues
- All pivots are positive
- All ^{sub} $\det A = +ve$

eg $\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$

$$(5-\lambda)(3-\lambda) - 4 = 0$$

$$\lambda^2 - 8\lambda + 15 - 4 = 0$$

$$\lambda^2 - 8\lambda + 11 = 0$$

$$\frac{8 \pm 2\sqrt{5}}{2} = \boxed{4 \pm \sqrt{5}}$$

